

Mohammad Rakibul Hasan Rifat

sajibahmedrifat1234@gmail.com | 01999984590 | [linkedin.com/in/rifat-923a14241](https://www.linkedin.com/in/rifat-923a14241) | github.com/Rifat4018 | Dhaka, Bangladesh

PROFESSIONAL SUMMARY

Results-driven Computer Science undergraduate (CGPA: 3.93/4.00) with hands-on research experience in Machine Learning, Computer Vision, NLP, and AI applied to biotech. Proficient in Python and PyTorch with a proven ability to implement scalable models, conduct baseline comparisons, and evaluate complex experiments. Experienced in benchmarking, reproducibility efforts, and preparing manuscripts for journal submission. Seeking a Research Internship to contribute technical expertise to cutting-edge AI/ML research and co-author high-impact publications.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, C#, JavaScript, HTML, CSS

Machine Learning & AI: Supervised/Unsupervised Learning, Federated Learning, GAN, GNN, NLP, Computer Vision, RAG, LLMs

Frameworks & Libraries: PyTorch, LangChain, FastAPI, FAISS, Scikit-learn, Pandas, NumPy

Tools & Platforms: Ollama, Google Colab, Kaggle, Git, VS Code, LaTeX, Linux

Software Engineering: OOP, Clean Code Principles, Agile Development, Unit Testing, System Design

EDUCATION

American International University Bangladesh (AIUB-Thesis)

Bachelor of Science in Computer Science and Engineering

Dhaka, Bangladesh

May 2022 – May 2026

- **CGPA: 3.93 / 4.00** | **Dean's Award List: 5 times**

- Major in Information Technology: Data Science, Data Mining & Warehousing, Advanced Database, Computer Vision

- Thesis: "Federated Learning and Graph Neural Network for Antimicrobial Resistance Prediction"

Adamjee Cantonment College

Higher Secondary Certificate (Science) – GPA: 5.00 / 5.00

Dhaka, Bangladesh

March 2018 – March 2020

Mirpur Cantonment Public School & College

Secondary School Certificate (Science) – GPA: 5.00 / 5.00

Dhaka, Bangladesh

March 2015 – March 2017

RESEARCH EXPERIENCE & PUBLICATIONS

Federated GNN (FedSSP-VRA) – Thesis Research | *AIUB*

Nov 2025 – Present

- Researched and implemented a personalized federated graph learning framework for multi-view genomic graph representations and AMR prediction using PyTorch.
- Conducted leakage-controlled benchmarking, ablation studies, and Explainable AI on graph structure design.

Object Detection Model Benchmarking | *Independent Research*

Feb 2026 – Apr 2026

- Benchmarked YOLOv8, YOLO11, Vision-Language Models(VLM) and RT-DETR models on satellite imagery datasets.
- Performed performance evaluation using mAP, precision, recall, and SAHI-based inference. Showing how model predict differently for small and big object.

3D Image Pancreas Radiomics Feature Extraction | *Python, Radiomics, Medical Imaging* Nov 2025 – Dec 2025

- Developed a radiomics-based medical imaging pipeline for feature extraction, clinical machine learning, and Explainable AI analysis from pancreatic imaging data.
- Accepted for presentation at [ICEFronT 2026](#).

Image Generation with GANs (SinGAN) | *Computer Vision*

Feb 2025 – Mar 2025

- Implemented GAN-based model for single-image synthesis and generation.

PROJECTS

- RAG-DocQA System** | *FastAPI, FAISS, Sentence Transformers, Ollama* May 2026
- Architected a Retrieval-Augmented Generation (RAG) backend utilizing sentence-transformers and FAISS for vector storage and semantic search across PDF and DOCX files.
 - Integrated local LLM inference via Ollama (Llama3) to retrieve relevant context and generate accurate, text-grounded answers.
- AI CV Screening & Job Matching Assistant** | *LangChain, FastAPI, Ollama* May 2026
- Engineered an AI-powered parsing pipeline using PyMuPDF and LangChain to dynamically extract, clean, and map candidate technical skills against targeted job descriptions.
 - Developed a modular REST API with FastAPI and Pydantic to automate skill gap analysis and generate foundational interview questions using local LLMs (phi3).
- BBC News NLP Pipeline & Titanic EDA** | *R, NLP, Data Science* Sep 2025 – Dec 2025
- Built an end-to-end **NLP pipeline** (scraping 250 articles, sentiment analysis, LDA topic modeling) and conducted comprehensive survival analysis with advanced feature engineering.

KEY ACHIEVEMENTS

- **Academic Excellence:** Recipient of Dean's Award List 5 consecutive times at AIUB (CGPA: 3.93/4.00).
- **Research Output:** Primary author and collaborator on AI/ML research encompassing Federated Learning, Graph Neural Networks, object detection benchmarking, and genomic data.
- **Problem Solving:** Participated in Hack Me If You Can (MIST LEETCON) and AI Reinventing events, demonstrating comfort with ambiguous algorithmic challenges.

ADDITIONAL INFORMATION

Languages: Bengali (Native), English (IELTS 6.5 – Professional Proficiency), Arabic (Reading), Hindi (Speaking)
Activities: Shoutcaster, AI community participant, hackathon competitor
University Email: 22-47371-2@student.aiub.edu | **Institution:** American International University Bangladesh

REFERENCES

Dipta Justin Gomes

Assistant professor, Faculty, Department of Computer Science
American International University – Bangladesh (AIUB)
Undergraduate Thesis Supervisor
Email: diptagomes@aiub.edu

Abhijit Kumar Ghosh

Lecturer, Department of Computer Science and Engineering (CSE)
European University of Bangladesh (EUB)
Email: abhijit@eub.edu.bd